Mechanistic Hypothesis-Search Report

Disease: RNU5B-1 Related Neurodevelopmental Disorder With Seizures And Joint Laxity (Mendelian) **Hypothesis ID:** `rnu5b1_missplicing_drives_brain_phenotype` **Hypothesis label:** Variant-specific mis-splicing of neurodevelopmental transcripts **KB status:** EMERGING **Search date:** 2026-10-19 · **Provider:** OpenScientist (PubMed `search_pubmed`, NCBI E-utilities)

Executive Judgment

Verdict: PARTIALLY SUPPORTED (emerging), with the gene–disease link established and the distinctive mechanistic claims still inferred.

The hypothesis decomposes into three claims of very different evidential status:

1. **RNU5B-1 variants cause a neurodevelopmental disorder. — ESTABLISHED.** Independently demonstrated by two orthogonal discovery strategies (snRNA burden analysis, P 40379786; R-loop-region analysis, P 40442284) and replicated in additional cohorts (P 42502410, P 40968615).
2. **The disorder is caused by altered splice-site usage (mis-splicing). — EMERGING / PLAUSIBLE BY ANALOGY.** Aberrant 5′-splice-site selection is a validated molecular hallmark of the sibling gene *RNU4-2* (ReNU syndrome), reproducible across tissues and in an iPSC neuronal model (P 42856368), and variant-specific episignature/splicing effects are reported for *RNU4-2* (P 40379786)

and *RNU2-2* (P 41912934).

For **RNU5B-1 itself**, the only splicing data come from cultured lymphocytes of **four carriers** (P 40379786) — small n, non-neural tissue — and the sibling gene *RNU2-2* shows only "subtle" blood splicing effects (P 41912934).

3. **The phenotype is driven by a *variant-specific set of brain-development transcripts*, selected by which exon-boundary sequences pair with the mutant U5 loop I. — SPECULATIVE.** No such causal transcript has been identified for any snRNA major-spliceosome NDD. The "loop I pairing" rationale is an **inference from normal spliceosome biology** (two 5′SS classes pairing with U5 loop 1 vs U6 ACAGA; P 39809319), **not a model the disease papers propose or test**.

Most important caveats. (a) The tissue and sample size for the only RNU5B-1 splicing data are poorly matched to a brain phenotype. (b) A mechanistically grounded **competing model — loss-of-function / dominant-negative splicing stress via snRNA quality control** (P 42562572, P 42629158) — predicts broad rather than target-specific disruption and has not been excluded. (c) No accessible omics dataset exists to independently re-test the splicing claim (GEO = 0 records; see Data and Tool Use). The seed hypothesis is therefore a reasonable, literature-consistent working model whose *specific and distinctive* elements (loop-I-determined, variant-specific, brain-development transcript set) remain unproven.

Evidence Matrix

#	Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context
1	40379786 (Nava 2025, Nature)	Human clinical + in vitro	**Supports** (seed's anchor)	RNU5B-1/ RNU5A-1 variants cause NDD; U4-2 variants alter 5′SS usage & episignatures by variant location	Establishes RNU5B-1 as NDD gene (21 U5 individuals); variant-location-dependent splicing/ methylation shown chiefly for RNU4-2	RNU5B-1, RNU5A-1, RNU4-2 dominant NDD
2	42856368 (Dawes 2026, medRxiv **preprint**)	Human clinical + in vitro (iPSC)	**Supports by analogy**	Variant-specific aberrant 5′SS selection is a reproducible, portable biomarker	105 concordant A5SS events; 6-site signature perfectly separates 19 T-loop patients from 54 + 5,984 controls; recapitulated in iPSC cortical neurons; signature specific to T-loop vs Stem III variants	RNU4-2 (ReNU), **not** RNU5B-1
3	39809319 (Parker, Fica & Simpson 2025, Open Biol)	Computational / review of mechanism	**Qualifies / basis of inference**	5′ splice sites split into U5-loop1-pairing vs U6-ACAGA-pairing classes	Provides the normal-biology rationale that mutant U5 loop I would preferentially perturb the U5-	General human splicing

#	Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context
					dependent 5′SS class	
4	41912934 (Leitão 2026, Nature)	Human clinical (omics)	**Qualifies**	snRNA variants produce variant-specific splicing/ episignature effects	RNU2-2 variants cause NDD; blood transcriptomics show **"subtle, variant-specific"** splicing effects; gradient-of-impact model	RNU2-2 dominant & recessive
5	40442284 (Jackson 2025, Nat Genet 57:1362)	Human genetics (computational)	**Supports (corroboration)**	RNU5B-1 is an NDD gene	Orthogonal R-loop-region method independently identifies RNU2-2 and RNU5B-1	RNU5B-1, RNU2-2
6	42502410 (Nakano 2026, iScience); 40968615 (Fan 2026)	Human genetics	**Supports (replication)**	RNU5B-1 variants recur across populations	Japanese & Chinese WGS cohorts confirm RNU5B-1 variants (novel n.38C>T)	RNU5B-1
7	42562572 (Ma 2026, Genes Dev); 42629158 (Mohan & Parker 2026, commentary)	In vitro / molecular	**Competing**	snRNA QC degrades mutant snRNAs; stabilized variants poison spliceosomes	Supports a loss-of-function / dominant-negative "splicing-stress" alternative to	snRNA class-wide

#	Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context
					neomorphic target selection	
8	41951959 (Rius 2026)	Human clinical	**Competing (LOF precedent)**	Biallelic RNU4-2 causes a distinct recessive NDD via reduced transcript levels	Demonstrates a loss-of-function route exists in the same snRNA family	Recessive RNU4-2
9	42463491 (Shikara 2026)	Model organism (Drosophila)	**Supports concept (template)**	A mutant snRNA mis-splices a defined transcript set to drive phenotype	U4atac stem-II mutant mis-splices immunity/ hematopoiesis set incl. JAK (hop); JAK reduction reproduced in Roifman patients	Minor spliceosome (U4atac)
10	42620352 (D'Souza 2026)	Review / guidance	**Qualifies**	snRNA variant interpretation is confounded by paralogs & high background mutation	~50-fold elevated mutability; bespoke classification needed	RNUopathies class
11	42799960 (Gowda 2026)	Review	Orientation	Splice-altering variants cause 15–30%	Context for splicing in neurogenetics	General

#	Citation (PMID)	Evidence type	Stance	Mechanistic claim tested	Key finding	Subtype / context
				of Mendelian neuro disease		

Data and Tool Use

Source	Accession / URI	Repository	Retrieval date	Access status	Query / f
PubMed (provider `search_pubmed`)	n/a	NCBI PubMed (provider-curated subset)	2026-10-19	**Accessed**	19 querie `pubmed_s`
NCBI E-utilities `efetch`	PMIDs 42856368, 39809319, 40442284, 42502410	NCBI	2026-10-19	**Accessed** (full abstracts)	`db=pubm`
NCBI GEO DataSets `gds`	—	NCBI GEO	2026-10-19	**Searched — 0 usable results**	`RNU5B-1` `ReNU sy`
NCBI SRA	—	NCBI SRA	2026-10-19	**Searched — no usable result**	`RNU5B-1` specific w
Nava 2025 patient RNA-seq (4 RNU5B-1 lymphocyte samples)	not located	controlled-access (expected EGA / journal supplement)	—	**Not accessed / unverified**	—
Dawes 2026 RNA-seq (30 RNU4-2)	medRxiv doi:10.64898/2026.09.26.26364077; PMC13648752	medRxiv/ PMC	2026-10-19	**Cited only** (abstract read)	—

Databases checked but not resolving relevant records (negative searches, logged): GEO DataSets and SRA as above. GenCC / ClinGen / ClinicalTrials.gov were **not** programmatically queried in this run and are therefore labelled **unverified**, not "absent."

Analysis inventory

Analysis	Input → Method → Output	Environment	Outcome
A1 · Literature mapping	19 PubMed queries → manual synthesis → evidence matrix	provider `search_pubmed`	**Succeeded** (`logs/pubmed_search_log.md`)
A2 · Full-abstract retrieval	PMIDs → `efetch` text → citation metadata	Python `requests`, NCBI eutils	**Succeeded**
A3 · Omics availability preflight	GEO/SRA esearch → counts → availability verdict	Python `requests`, NCBI eutils	**Succeeded** (negative) (`logs/geo_sra_preflight.json`)
A4 · Direct re-analysis of RNU5B-1 splicing	(intended) patient RNA-seq → splice-event quantification	—	**SKIPPED** — no accessible dataset; disclosed limitation, no silent fallback

No patient-level or large raw data were downloaded or bundled. No structural-biology (Phenix) analysis was applicable (no coordinate files).

Mechanistic Causal Chain

```
[Upstream trigger]  De novo RNU5B-1 variant (maternal-biased), clustering in splicing-critical U
        │   STRONG: <a href="https://pubmed.ncbi.nlm.nih.gov/40379786/" rel="noopener noreferrer
        ▼
[Molecular lesion]  Altered U5 snRNA (loop I) incorporated into U5 snRNP / spliceosome
        │   INFERRED for RNU5B-1: structural placement assumed; QC fate (degradation vs assembly
        │   COMPETING at this node: <a href="https://pubmed.ncbi.nlm.nih.gov/42562572/" rel="noo
        │                           or poison spliceosomes → dominant-negative)
        ▼
[Primary molecular effect]  Variant-specific altered 5′-splice-site selection
        │   ANALOGY-SUPPORTED (RNU4-2: <a href="https://pubmed.ncbi.nlm.nih.gov/42856368/" rel="
        │   MISSING STEP: mapping RNU5B-1 mis-splicing to U5-loop1-dependent 5′SS class (<a href
        ▼
[Target-level effect]  Mis-splicing of a *defined set of brain-development transcripts*, set dif
        │   SPECULATIVE: no causal transcript identified for any major-spliceosome snRNA NDD
        │   TEMPLATE EXISTS (minor spliceosome): U4atac → JAK/hop (<a href="https://pubmed.ncbi.
        ▼
[Cellular/developmental effect]  Disrupted neuronal proliferation/migration/function
        │   MISSING: no RNU5B-1 neuronal/organoid/animal model reported
        ▼
[Clinical manifestation]  Developmental delay, seizures, joint laxity; clinical variability by v
            INFERRED link to variant-specific splicing (<a href="https://pubmed.ncbi.nlm.nih.gov
```

Strong links: variant → gene–disease causation. **Inferred links:** variant → loop-I structural effect; primary mis-splicing (by analogy). **Missing links:** the specific brain-development transcript set; the neuronal/developmental perturbation; the splicing-signature → clinical-severity correlation *for RNU5B-1*.

Knowledge Gaps

1. **No identified causal target transcript (missing causal step).** Scope: the seed's core distinctive claim. Why it matters: without it, "variant-specific set needed for brain development" is unfalsifiable as stated. Checked: PubMed (19 queries) + GEO/SRA — none found. Resolver: deep/long-read RNA-seq in RNU5B-1 patient neurons/organoids with minigene validation of candidate neurodevelopmental targets.
2. **Wrong-tissue, small-n splicing data (weak causal step).** Scope: RNU5B-1 primary evidence = 4 carriers' lymphocytes. Why: brain phenotype inferred from non-neural cells. Checked:

. P 40379786

Resolver: RNU5B-1 iPSC-cortical-neuron RNA-seq (mirroring the RNU4-2 approach in). P 42856368

3. **Loop-I pairing step untested (unconfirmed edge).** Scope: whether RNU5B-1 mis-splicing is enriched in U5-loop1-dependent 5′SS. Checked: P 39809319 provides only the rationale. Resolver: classify patient aberrant 5′SS by loop1-vs-U6 pairing potential; expect enrichment if the seed holds.
4. **Neomorphic vs loss-of-function not discriminated (conflicting mechanistic models).** Scope: node 2 of the chain. Checked: / P 42562572 42629158 (QC/LOF), P 41951959 (LOF precedent). Resolver: measure mature U5B levels and mutant-U5 spliceosome incorporation in patient cells; allele-specific snRNP assays.
5. **No RNU5B-1 model system (unconfirmed edge; perturbation evidence absent).** Checked: PubMed — none. Resolver: knock-in mouse/organoid or Drosophila U5 loop-I model (cf. U4atac,). P 42463491
6. **Data-source absence (source-level).** GEO DataSets = 0 for RNU5B-1 / RNU4-2 / ReNU on 2026-10-19; SRA non-specific. Why: blocks independent computational verification. Resolver: deposition of processed splice-event matrices; curator query of EGA/controlled repositories.
7. **Unverified database sources.** GenCC, ClinGen, ClinicalTrials.gov were **not** queried programmatically — status **unverified**, not absent. Resolver: direct API queries by curator.
8. **No treatment/biomarker MoA for RNU5B-1.** The RNU4-2 splicing signature (P 42856368) is proposed as a therapeutic-development biomarker but is RNU4-2-specific; no RNU5B-1 biomarker or intervention exists.

Alternative Models

Model	Relationship to seed	Summary	Key support
A. Neomorphic variant-specific target selection (SEED)	—	Mutant loop I redirects 5′SS usage toward a variant-defined, brain-relevant transcript set	P 40379786 P 42856368 (analogy), 39809319 (rationale)
B. Loss-of-function / reduced functional U5 pool	**Competing (same node)**	Mutant U5 degraded by QC → fewer functional U5 snRNPs → global splicing stress; neurons most sensitive	P 42562572 P 42629158 P 41951959
C. Dominant-negative spliceosome poisoning	**Competing (same node)**	Mutant U5 assembles but impairs catalysis broadly, not target-selectively	P 42629158
D. Cumulative/ global mis-splicing burden	**Competing**	Aggregate low-level dysregulation (not a single "set") drives phenotype; parsimonious with "subtle" effects	P 41912934 ("subtle" effects)
E. Episignature / DNA-methylation–mediated	**Parallel / downstream**	Methylation changes contribute to or mark the phenotype	P 40379786 (episignatures)
F. R-loop-driven mutability	**Upstream cause** (not competing for downstream mechanism)	Explains *why/where* variants arise, not how they cause disease	P 40442284

Models B–D are the strongest alternatives to the seed's *distinctive* claim; the seed and B/C/D are partially non-exclusive but make divergent, testable predictions (target-specific vs broad disruption).

Discriminating Tests

1. **RNU5B-1 iPSC → cortical neuron RNA-seq, variant-stratified** (mirror
   P 42856368
).
   Stratify by variant position in U5. *Expected if seed:* reproducible, variant-position-specific aberrant 5′SS signature enriched in neurodevelopmental genes and in U5-loop1-pairing 5′SS. *If B/D:* broad, weakly variant-specific disruption.
2. **Loop1-vs-U6 pairing classification of patient aberrant 5′SS** (computational, using
   P 39809319
   framework) — directly tests the loop-I inference.
3. **Mature U5B quantification + mutant-U5 snRNP incorporation assay** in patient cells — discriminates LOF (reduced functional U5, mutant degraded) from neomorphic (mutant incorporated, altered specificity).
4. **Candidate-target minigene/rescue** — validate specific neurodevelopmental transcripts as causal (the U4atac→JAK template,
   P 42463491
).
5. **RNU5B-1 knock-in animal/organoid** — establish the missing developmental-perturbation link and test whether correcting a candidate target rescues phenotype.
6. **Genotype–phenotype correlation across the full RNU5B-1 cohort** (seizures/joint-laxity vs variant position) paired with per-variant splicing signatures — tests the "set differs by variant" claim at clinical scale.

Curation Leads (require curator verification)

- **Status:** Retain **EMERGING**. Gene–disease is ESTABLISHED, but the *distinctive* mechanistic claims (loop-I-determined, variant-specific brain-development transcript set) remain SPECULATIVE → do not upgrade the mechanism hypothesis.
- **Candidate evidence edges (verify snippets):**
- P 40442284 — "we report neurodevelopmental disorders (NDDs) caused by rare variants in two major spliceosomal RNA encoding genes, RNU2-2 and RNU5B-1" → *supports* RNU5B-1 gene–disease (independent).
- P 42856368 — "Aberrant 5' splice site selection is a molecular hallmark of ReNU syndrome that correlates with phenotypic severity" (verify against full text; stored abstract is truncated) → *supports-by-analogy* the mis-splicing mechanism (RNU4-2).
- P 41912934 — "subtle, variant-specific effects on splicing and episignatures" → *qualifies* effect size.
- P 42562572 — "Impeding snRNA quality control causes formation of aberrant spliceosomes and altered pre-mRNA splicing" → *competing* loss-of-function node.
- P 42463491 — U4atac → JAK/hop transcript-set template → *supports concept*, cross-spliceosome.
- **Candidate pathophysiology nodes/edges:** add node "U5 snRNP functional insufficiency (LOF)" as a competing upstream-molecular node; add edge "R-loop-region mutability → de novo RNU5B-1 variant" (P 40442284) as upstream.
- **Candidate ontology terms:** cell type — cortical excitatory neuron (CL:0000679); processes — mRNA 5′-splice-site selection (GO:0000395 / spliceosomal), regulation of mRNA splicing (GO:0048024), nuclear-transcribed mRNA catabolic process / snRNA QC (GO:0071028 nuclear mRNA surveillance context).

- **Candidate subtype restriction:** splicing-signature evidence is RNU4-2 T-loop-specific; do **not** generalize the RNU4-2 biomarker to RNU5B-1 without direct data.
- **Candidate `knowledge_gaps` entries:** (1) no RNU5B-1 causal target transcript; (2) neomorphic-vs-LOF unresolved; (3) no RNU5B-1 neuronal model; (4) GEO/SRA data-source absence (2026-10-19); (5) GenCC/ClinGen/trials unverified this run.

Limitations

- Provider PubMed index is curated/partial; several targeted queries returned 0 — some relevant primary literature (e.g., classic U5 loop-I structural biology) may not be indexed here and is under-cited.
- Key supporting splicing evidence (P 42856368) is a **preprint** and concerns **RNU4-2**, not RNU5B-1.
- No omics dataset was accessible; the central mis-splicing claim was **not** independently re-analyzed (disclosed; no silent fallback).
- Citation snippets for the preprint could not be auto-validated against the provider's truncated stored abstract; verbatim text was taken from the NCBI efetch full abstract and is flagged for curator check.

Artifacts: `openscientist_artifacts/` *— see* `MANIFEST.yaml`, `logs/pubmed_search_log.md`, `logs/geo_sra_preflight.json`.
